Algorithm 1 Meta-evolution

Inputs: baseline; G_{total} total generations; G_{simplify} simplification generations; O offspring; P population size; training tasks T; validation tasks T_{val}; F fitness metric; N_{init} initial evaluation seeds; N_{val} validation seeds; reevaluation `strategy`: either `TTTS` with B seeds per generation, or `uniform` with N_{reeval} extra seeds per new population member.

`population` $\leftarrow$ [`baseline`]
`archive` $\leftarrow \{\}$
`new_members` $\leftarrow \{\}$
for $i = 0, \ldots, G_{\text{total}} - 1$ **do**
  # Generate offspring
  `offspring` $\leftarrow \{\}$
  for $o = 0, \ldots, O - 1$ **do**
    `sr_operator` $\leftarrow$ RandomChoice([`mutation`, `survival`, `selection`, `loss`])
    if $i = 0$ **then**
      `mode` $\leftarrow$ `explore`
    else if $G_{\text{total}} - i \leq G_{\text{simplify}}$ **then**
      `mode` $\leftarrow$ `simplify`
    else
      `mode` $\leftarrow$ RandomChoice([`explore`, `refine`, `crossover`, `simplify`])
    end if
    `parent` $\leftarrow$ TournamentSelection(`population`, $n = 2$)
    if `mode` $=$ `crossover` **then**
      `parent2` $\leftarrow$ TournamentSelection(`population`, $n = 2$)
    end if
    `feedback` $\leftarrow$ GenerateTraceFeedback(T, `parent`)
    `prompt` $\leftarrow$ GeneratePrompt(`sr_operator`, `mode`, `parent`, `feedback`, `parent2`)
    repeat
      `x` $\leftarrow$ `LLM_ensemble.sample().sample_completion(prompt)`
    until PassesValidation(`x`)
    `offspring` $\leftarrow$ `offspring` $\cup \{$`x`$\}$
  end for
    # Choose reevaluations
  if `strategy` $=$ `TTTS` **then**
    $\psi_x \leftarrow$ top-two Thompson probability of x under $\mathcal{N}(\hat{\mu}_x, \hat{\sigma}^2/n_x)$, for $x \in$ `archive`
    `reeval` $\leftarrow B$ draws from `archive` with probability $\propto \psi_x$
  else
    `reeval` $\leftarrow \{x : N_{\text{reeval}} \mid x \in$ `new_members`$\}$
  end if
    # Parallel evaluation of offspring and reevaluations
  `evaluation_batch` $\leftarrow \{x : N_{\text{init}} \mid x \in$ `offspring`$\} \cup$ `reeval`
  ParallelEvaluation(`evaluation_batch`, T, F)
    # Update archive of scores, averaged over all seeds and tasks so far
  for $x \in$ `evaluation_batch` **do**
    $\hat{\mu}_x \leftarrow$ mean of $F(x, \text{task}, \text{seed})$ over all evaluated tasks and seeds
    $n_x \leftarrow$ number of seeds evaluated
    `archive` $\leftarrow$ `archive` $\cup \{x\}$
  end for
  if $G_{\text{total}} - i \leq G_{\text{simplify}}$ **then** # LOC versus score population
    Form the Pareto frontier of score versus LOC.
    Fill remaining slots (up to P) with individuals selected by highest overall score.
  else # Task-based population
    For each task, add the individual with highest average score on that task to population.
    Fill remaining slots (up to P) with individuals selected by highest overall score.
  end if
  `new_members` $\leftarrow \{x \in$ `offspring` $\mid x \in$ `population`$\}$
    # Validate the generation's best on held-out tasks
  $x_i^* \leftarrow \arg\max_{x \in \texttt{population}} \hat{\mu}_x$
  `val`$[x_i^*] \leftarrow \frac{1}{N_{\text{val}}|T_{\text{val}}|} \sum_{\text{seeds, task} \in T_{\text{val}}} F(x_i^*, \text{task}, \text{seed})$
end for
`best_individual` $\leftarrow \arg\max_x$ `val`$[x]$
return `best_individual`